

MET CS662 - Assignment #8

1. Design a Turing Machine with no more than three states that accepts the language $L(a(a + b)^*)$. Assume that $\Sigma = \{a, b\}$. Is it possible to do this with a two-state machine?

$$M = (Q, \Sigma, \Gamma, \delta, q_0, \diamond, F)$$

$$Q = \{q_0, q_1\}$$

$$\Sigma = \{a, b\}$$

$$\Gamma = \{a, b, \diamond\}$$

$$\delta(q_0, a) = (q_1, a, R)$$

$$F = \{q_1\}$$

2. What language is accepted by the TM, $M = (\{q_0, q_1, q_2, q_3\}, \{a, b\}, \{a, b, \square\}, \delta, q_0, \square, \{q_3\})$ with

$$\delta(q_0, a) = (q_1, a, R) \quad \delta(q_1, \square) = (q_3, \square, R)$$

$$\delta(q_0, b) = (q_2, b, R) \quad \delta(q_2, b) = (q_2, b, R)$$

$$\delta(q_1, b) = (q_1, b, R) \quad \delta(q_2, a) = (q_3, a, R)$$

$$L(M) = \{ab^n \mid n \geq 0\} \cup \{b^m a \omega \mid m \geq 1, \omega \in \{a, b\}^*\}$$

$$L(M) = (ab^* + bb^*a(a + b)^*)$$

3. What happens in Example 9.10 (in the textbook where the TM copies strings of 1's, i.e. the TM that performs $q_0 \omega \vdash^* q_f \omega \omega$) if the string ω contains any symbol other than 1?

The Turing Machine will halt if it encounters a symbol other than 1, x , or $\square$, as all the transitions are only defined for input symbols 1, x , and $\square$. It will reject any string containing a symbol other than 1.

4. Outline with enough details a TM that accepts the language $L = \{\omega \omega \mid \omega \in \{a, b\}^+\}$.
- Replace the leftmost a or b . Replace a with u and b with v .
 - Replace the rightmost a or b . Replace a with x and b with y .
 - Repeat steps (i) and (ii) until no more a s or b s remain. Now, the first copy of ω has all a and b replaced with u and v , respectively. The second copy of ω has all a and b replaced with x and y , respectively. If the string is odd, there will be an extra u or v , so halt and reject.
 - Replace the leftmost symbol that is u or v .
 - If the leftmost symbol is u , replace it with z then find the leftmost symbol that is x or y . If it is x , replace it with z . If it is y or there are no x or y remaining, halt and reject the string.
 - If the leftmost symbol is v , replace it with z then find the leftmost symbol that is x or y . If it is y , replace it with z . If it is x or there are no x or y remaining, halt and reject the string.
 - Repeat step (iv) until no more u or v remain. Once every u and v has been matched with the corresponding x and y , accept the string.

$$M = (Q, \Sigma, \Gamma, \delta, q_0, \diamond, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}\}$$

$$\Sigma = \{a, b\}$$

$$\Gamma = \{a, b, u, v, x, y, z, \diamond\}$$

$$F = \{q_{10}\}$$

State 0: Replace leftmost a/b with u/v

$$\delta(q_0, a) = (q_1, u, R)$$

$$\delta(q_0, b) = (q_1, v, R)$$

$$\delta(q_0, x) = (q_4, x, L)$$

$$\delta(q_0, y) = (q_4, y, L)$$

State 1: Find rightmost a/b

$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, b) = (q_1, b, R)$$

$$\delta(q_1, \diamond) = (q_2, \diamond, L)$$

$$\delta(q_1, x) = (q_2, x, L)$$

$$\delta(q_1, y) = (q_2, y, L)$$

State 2: Replace rightmost a/b with x/y

$$\delta(q_2, a) = (q_3, x, L)$$

$$\delta(q_2, b) = (q_3, y, L)$$

Transitions for u and v in state 2 are not defined because they will only be encountered in state 2 if the input string is odd length and should be rejected.

State 3: Find leftmost a/b

$$\delta(q_3, a) = (q_3, a, L)$$

$$\delta(q_3, b) = (q_3, b, L)$$

$$\delta(q_3, u) = (q_0, u, R)$$

$$\delta(q_3, v) = (q_0, v, R)$$

State 4: Scan to left of string

$$\delta(q_4, u) = (q_4, u, L)$$

$$\delta(q_4, v) = (q_4, v, L)$$

$$\delta(q_4, \diamond) = (q_5, \diamond, R)$$

State 5: Mark leftmost u/v

$$\delta(q_5, u) = (q_6, z, R)$$

$$\delta(q_5, v) = (q_7, z, R)$$

State 6: Mark matching x

$$\delta(q_6, u) = (q_6, u, R)$$

$$\delta(q_6, v) = (q_6, v, R)$$

$$\delta(q_6, z) = (q_6, z, R)$$

$$\delta(q_6, x) = (q_8, z, L)$$

Halt and reject if y or $\diamond$ is read. Input string is not of the form $\omega\omega$.

State 7: Mark matching y

$$\delta(q_7, u) = (q_7, u, R)$$

$$\delta(q_7, v) = (q_7, v, R)$$

$$\delta(q_7, z) = (q_7, z, R)$$

$$\delta(q_7, y) = (q_8, z, L)$$

Halt and reject if x or $\diamond$ is read. Input string is not of the form $\omega\omega$.

State 8: Scan to middle of string

$$\delta(q_8, z) = (q_8, z, L)$$

$$\delta(q_8, u) = (q_9, u, L)$$

$$\delta(q_8, v) = (q_9, v, L)$$

$$\delta(q_8, \diamond) = (q_{10}, \diamond, R)$$

If $\diamond$ is read, all u and v have been successfully matched, so accept the string.

State 9: Find leftmost u/v

$$\delta(q_9, u) = (q_9, u, L)$$

$$\delta(q_9, v) = (q_9, v, L)$$

$$\delta(q_9, z) = (q_5, z, R)$$

State 10: Final state
